In [1]:

```
import pandas as pd
import numpy as np
df = pd.read_csv("Reviews.csv")
print("Q1: Dataset Loaded Successfully")
display(df.head())

!pip install nltk scikit-learn -q
import re
import string
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
print("\nQ2: Libraries Imported Successfully")
# Q3: Download required NLTK resources

nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')
print("\nQ3: NLTK Resources Downloaded")

# Q4: Clean the dataset
df = df[['Text']].dropna()

# Remove duplicate reviews
df = df.drop_duplicates(subset=['Text'])

# Retain only the first 10,000 reviews
df = df.head(10000).copy()
df.reset_index(drop=True, inplace=True)
print("\nQ4: Dataset Cleaning")
print("Number of reviews:", len(df))
print("Missing values:", df['Text'].isnull().sum())
print("Duplicate reviews:", df['Text'].duplicated().sum())

# Q5: Create a text preprocessing function

stop_words = set(stopwords.words('english'))
def preprocess_text(text):
    text = text.lower()
    text = re.sub(r'[^a-z\s]', ' ', text)
    tokens = word_tokenize(text)
    tokens = [word for word in tokens if word not in stop_words]
    cleaned_text = ' '.join(tokens)
    return cleaned_text

print("\nQ5: Preprocessing Function Created")

# Q6: Apply preprocessing to all reviews

df['Cleaned_Text'] = df['Text'].apply(preprocess_text)

print("\nQ6: Preprocessing Completed")
display(df.head())

# Q7: Display original and cleaned versions of five reviews
```

```
print("\nQ7: Original and Cleaned Reviews")
for i in range(min(5, len(df))):
    print(f"\nReview {i + 1}")
    print("Original Review:", df.loc[i, 'Text'])
    print("Cleaned Review:", df.loc[i, 'Cleaned_Text'])


# Q8: Convert cleaned reviews into TF-IDF feature vectors

vectorizer = TfidfVectorizer()
tfidf_matrix = vectorizer.fit_transform(df['Cleaned_Text'])

print("\nQ8: TF-IDF Vectorization")
print("TF-IDF Matrix Shape:", tfidf_matrix.shape)
print("Vocabulary Size:", len(vectorizer.vocabulary_))

# Q9: Information Retrieval using Cosine Similarity

def retrieve_reviews(query, top_n=5):
    # Preprocess the user query
    cleaned_query = preprocess_text(query)

    # Convert the query into a TF-IDF vector
    query_vector = vectorizer.transform([cleaned_query])

    # Calculate cosine similarity with all reviews
    similarities = cosine_similarity(
        query_vector, tfidf_matrix).flatten()

    # Retrieve the top matching reviews
    top_indices = np.argsort(similarities)[::-1][:top_n]

    print("\nSearch Query:", query)
    print("Cleaned Query:", cleaned_query)
    print("=" * 80)

    for rank, idx in enumerate(top_indices, start=1):
        print(f"\nRank: {rank}")
        print("Similarity Score:", round(similarities[idx], 4))
        print("Original Review:", df.iloc[idx]['Text'])
        print("Cleaned Review:", df.iloc[idx]['Cleaned_Text'])
        print("-" * 80)

# Q10: Test the retrieval function

retrieve_reviews("great product with fast shipping")
retrieve_reviews("disappointed")
retrieve_reviews("excellent taste")
retrieve_reviews("poor packaging")

# Q11: Compare retrieved reviews for different queries

queries = [
    "great product with fast shipping",
    "disappointed",
    "excellent taste",
    "poor packaging"
]

print("\nQ11: Query Evaluation")
```

```
for query in queries:
    cleaned_query = preprocess_text(query)
    query_vector = vectorizer.transform([cleaned_query])

    similarities = cosine_similarity(
        query_vector, tfidf_matrix
    ).flatten()

    top_indices = np.argsort(similarities)[::-1][:5]

    print("\nQuery:", query)

    for idx in top_indices:
        print(
            "Similarity:", round(similarities[idx], 4),
            "| Review:", df.iloc[idx]['Text'][:200])

# Q12: Relevance Analysis

print("\nQ12: Relevance Analysis")

print("1. 'great product with fast shipping': Retrieves reviews "
      "sharing terms related to product quality and shipping.")

print("2. 'disappointed': Retrieves reviews containing matching "
      "terms that may indicate customer dissatisfaction.")

print("3. 'excellent taste': Retrieves reviews containing terms "
      "related to taste and positive feedback.")

print("4. 'poor packaging': Retrieves reviews containing terms "
      "related to packaging and negative feedback.")
```

```
Q1: Dataset Loaded Successfully
```

	Id	ProductId	UserId	ProfileName	HelpfulnessNumerator	Helpfulness
0	1	B001E4KFG0	A3SGXH7AUHU8GW	delmartian	1	
1	2	B00813GRG4	A1D87F6ZCVE5NK	dll pa	0	
2	3	B000LQOCH0	ABXLMWJIXXAIN	Natalia Corres "Natalia Corres"	1	
3	4	B000UA0QIQ	A395BORC6FGVXV	Karl	3	
4	5	B006K2ZZ7K	A1UQRSCLF8GW1T	Michael D. Bigham "M. Wassir"	0	

```
Q2: Libraries Imported Successfully
[nltk_data] Downloading package punkt to
[nltk_data]     C:\Users\rena\AppData\Roaming\nltk_data...
[nltk_data]   Unzipping tokenizers\punkt.zip.
[nltk_data] Downloading package punkt_tab to
[nltk_data]     C:\Users\rena\AppData\Roaming\nltk_data...
[nltk_data]   Unzipping tokenizers\punkt_tab.zip.
[nltk_data] Downloading package stopwords to
[nltk_data]     C:\Users\rena\AppData\Roaming\nltk_data...
[nltk_data]   Unzipping corpora\stopwords.zip.
Q3: NLTK Resources Downloaded

Q4: Dataset Cleaning
Number of reviews: 10000
Missing values: 0
Duplicate reviews: 0

Q5: Preprocessing Function Created

Q6: Preprocessing Completed
```

	Text	Cleaned_Text
0	I have bought several of the Vitality canned d...	bought several vitality canned dog food produc...
1	Product arrived labeled as Jumbo Salted Peanut...	product arrived labeled jumbo salted peanuts p...
2	This is a confection that has been around a fe...	confection around centuries light pillowy citr...
3	If you are looking for the secret ingredient i...	looking secret ingredient robitussin believe f...
4	Great taffy at a great price. There was a wid...	great taffy great price wide assortment yummy ...

Q7: Original and Cleaned Reviews

Review 1
Original Review: I have bought several of the Vitality canned dog food products a
nd have found them all to be of good quality. The product looks more like a stew
than a processed meat and it smells better. My Labrador is finicky and she apprec
iates this product better than most.
Cleaned Review: bought several vitality canned dog food products found good quali
ty product looks like stew processed meat smells better labrador finicky apprecia
tes product better

Review 2
Original Review: Product arrived labeled as Jumbo Salted Peanuts...the peanuts we
re actually small sized unsalted. Not sure if this was an error or if the vendor
intended to represent the product as "Jumbo".
Cleaned Review: product arrived labeled jumbo salted peanuts peanuts actually sma
ll sized unsalted sure error vendor intended represent product jumbo

Review 3
Original Review: This is a confection that has been around a few centuries. It i
s a light, pillowy citrus gelatin with nuts - in this case Filberts. And it is cu
t into tiny squares and then liberally coated with powdered sugar. And it is a t
iny mouthful of heaven. Not too chewy, and very flavorful. I highly recommend t
his yummy treat. If you are familiar with the story of C.S. Lewis' "The Lion, Th
e Witch, and The Wardrobe" - this is the treat that seduces Edmund into selling o
ut his Brother and Sisters to the Witch.
Cleaned Review: confection around centuries light pillowy citrus gelatin nuts cas
e filberts cut tiny squares liberally coated powdered sugar tiny mouthful heaven
chewy flavorful highly recommend yummy treat familiar story c lewis lion witch wa
rdrobe treat seduces edmund selling brother sisters witch

Review 4
Original Review: If you are looking for the secret ingredient in Robitussin I bel
ieve I have found it. I got this in addition to the Root Beer Extract I ordered
(which was good) and made some cherry soda. The flavor is very medicinal.
Cleaned Review: looking secret ingredient robitussin believe found got addition r
oot beer extract ordered good made cherry soda flavor medicinal

Review 5
Original Review: Great taffy at a great price. There was a wide assortment of yu
mmy taffy. Delivery was very quick. If your a taffy lover, this is a deal.
Cleaned Review: great taffy great price wide assortment yummy taffy delivery quic
k taffy lover deal

Q8: TF-IDF Vectorization
TF-IDF Matrix Shape: (10000, 18521)
Vocabulary Size: 18521

Search Query: great product with fast shipping
Cleaned Query: great product fast shipping
==========

Rank: 1
Similarity Score: 0.5103
Original Review: Enjoyed the product and they also provided very fast shipping.
I'm about out and need to order more.
Cleaned Review: enjoyed product also provided fast shipping need order
----

Rank: 2

Similarity Score: 0.4925
Original Review: The tea is good and fresh. We enjoy it. The shipping is fast and cost is reasonable.
I recommend this product.
Cleaned Review: tea good fresh enjoy shipping fast cost reasonable br recommend p roduct
---

Rank: 3
Similarity Score: 0.4001
Original Review: These are very good, they were a great price and fast-free shipp ing when you purchase enough food. The only thing is that the box was smaller th an I guessed it would be.
Cleaned Review: good great price fast free shipping purchase enough food thing bo x smaller guessed would
---

Rank: 4
Similarity Score: 0.3996
Original Review: The energy drink is a great product. The shipping price is crazy tho. I paid more for the shipping then both products together, so not a great dea l.
Cleaned Review: energy drink great product shipping price crazy tho paid shipping products together great deal
---

Rank: 5
Similarity Score: 0.3936
Original Review: Was exactly what I was looking for. Super fast shipping! A bit o n the spicy side,but great taste!!!!
Would definately purchase them again!
Cleaned Review: exactly looking super fast shipping bit spicy side great taste br would definately purchase
---

Search Query: disappointed
Cleaned Query: disappointed
==========

Rank: 1
Similarity Score: 0.4846
Original Review: I am a bit disappointed. The flavor was not what I wanted or ex pected.
Cleaned Review: bit disappointed flavor wanted expected

Rank: 2
Similarity Score: 0.3582
Original Review: The product is very good. Way too expensive and at almost 5.00 a box to get pancakes or biscuits I was disappointed
Cleaned Review: product good way expensive almost box get pancakes biscuits disap pointed
---

Rank: 3
Similarity Score: 0.3216
Original Review: I was disappointed in this product because I thought it would be bigger.
Also, it did not come with enough icing. I had to use my own.
Cleaned Review: disappointed product thought would bigger br also come enough ici ng use
---

Rank: 4
Similarity Score: 0.3195
Original Review: Disappointed. The big boxes had a very different flavor than th
e smaller boxes. I am not sure why but not good.
Cleaned Review: disappointed big boxes different flavor smaller boxes sure good
----

Rank: 5
Similarity Score: 0.3117
Original Review: Just plain nasty!!! This item tasted like cardboard, very watere
d down and no flavor to it at all. Very disappointed
Cleaned Review: plain nasty item tasted like cardboard watered flavor disappointe
d

Search Query: excellent taste
Cleaned Query: excellent taste
==

Rank: 1
Similarity Score: 0.502
Original Review: It is very very good sweets
Amazon always is excellent<br /
>Amazon always is excellent
Amazon always is excellent
Amazon always is
excellent
Cleaned Review: good sweets br amazon always excellent br amazon always excellent
br amazon always excellent br amazon always excellent

Rank: 2
Similarity Score: 0.4605
Original Review: The Popchips were excellent - had excellent flavor and low in ca
lories. When you eat these - don't need another potato chip!
Cleaned Review: popchips excellent excellent flavor low calories eat need another
potato chip

Rank: 3
Similarity Score: 0.4054
Original Review: Excellent product. Exactly what they said it would be. Superior
chamomile from what I have ordered before from other companies. Excellent value f
or the price.
Cleaned Review: excellent product exactly said would superior chamomile ordered c
ompanies excellent value price

Rank: 4
Similarity Score: 0.3856
Original Review: We recieved the product on time and in excellent shape. It taste
s good and meets our expectations. Thanks for the excellent service.
Cleaned Review: recieved product time excellent shape tastes good meets expectati
ons thanks excellent service

Rank: 5
Similarity Score: 0.384
Original Review: Title says it all, a Excellent quality product and Excellent ser
vice at a fair price. Product came ahead of time and as expected.
Cleaned Review: title says excellent quality product excellent service fair price
product came ahead time expected

Search Query: poor packaging
Cleaned Query: poor packaging
==========

Rank: 1
Similarity Score: 0.3246
Original Review: I was delighted to find this product online, because I cannot al
ways find it in the stores. However, each package I have opened thus far is at l
east half reduced to crumbs because of poor packaging for shipping. I doubt that
I will order more using this method.
Cleaned Review: delighted find product online always find stores however package
opened thus far least half reduced crumbs poor packaging shipping doubt order usi
ng method

Rank: 2
Similarity Score: 0.3085
Original Review: This review is for the packaging and not the actual mix, since t
he mix is a very subjective thing and my taste my not match others and, some wrot
e about how poor the packaging was. All I can say is with mine there is more than
enough labeling, on the bags themselves and on the box there are instructions on
how to. Granted the box is nothing heavy duty or special and the mix is in plain
white bags, there are more than enough instructions to go by.
Cleaned Review: review packaging actual mix since mix subjective thing taste matc
h others wrote poor packaging say mine enough labeling bags box instructions gran
ted box nothing heavy duty special mix plain white bags enough instructions go
---

Rank: 3
Similarity Score: 0.3084
Original Review: The soup is rich and tasty, but DON'T buy it from Amazon. The p
ackaging is so poor that the cans arive dented as did the replacement shipment. I
got my money back but now have to pay more to buy this wonderful soup from my loc
al grocer.
Cleaned Review: soup rich tasty buy amazon packaging poor cans arive dented repla
cement shipment got money back pay buy wonderful soup local grocer
---

Rank: 4
Similarity Score: 0.292
Original Review: Didnt really enjoy these after wanting to try something other th
an aunt jemima's from Dominicks. The pancakes tasted strange and sometimes forget
table, but it might have been due to the poor packaging they came in along with t
he syrup.
Cleaned Review: didnt really enjoy wanting try something aunt jemima dominicks pa
ncakes tasted strange sometimes forgettable might due poor packaging came along s
yrup

Rank: 5
Similarity Score: 0.2811
Original Review: was really dissappointed in product... too salty and poor flavo
r....Convenience is what makes product worth purchasing if you like pork rinds...
Cleaned Review: really dissappointed product salty poor flavor convenience makes
product worth purchasing like pork rinds

Q11: Query Evaluation

Query: great product with fast shipping
Similarity: 0.5103 | Review: Enjoyed the product and they also provided very fast shipping. I'm about out and need to order more.
Similarity: 0.4925 | Review: The tea is good and fresh. We enjoy it. The shipping is fast and cost is reasonable.
I recommend this product.
Similarity: 0.4001 | Review: These are very good, they were a great price and fas t-free shipping when you purchase enough food. The only thing is that the box wa s smaller than I guessed it would be.
Similarity: 0.3996 | Review: The energy drink is a great product. The shipping pr ice is crazy tho. I paid more for the shipping then both products together, so no t a great deal.
Similarity: 0.3936 | Review: Was exactly what I was looking for. Super fast shipp ing! A bit on the spicy side,but great taste!!!!
Would definately purchase t hem again!

Query: disappointed
Similarity: 0.4846 | Review: I am a bit disappointed. The flavor was not what I wanted or expected.
Similarity: 0.3582 | Review: The product is very good. Way too expensive and at a lmost 5.00 a box to get pancakes or biscuits I was disappointed
Similarity: 0.3216 | Review: I was disappointed in this product because I thought it would be bigger.
Also, it did not come with enough icing. I had to use my own.
Similarity: 0.3195 | Review: Disappointed. The big boxes had a very different fl avor than the smaller boxes. I am not sure why but not good.
Similarity: 0.3117 | Review: Just plain nasty!!! This item tasted like cardboard, very watered down and no flavor to it at all. Very disappointed

Query: excellent taste
Similarity: 0.502 | Review: It is very very good sweets
Amazon always is exc ellent
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Amazo n always is excellent
Similarity: 0.4605 | Review: The Popchips were excellent - had excellent flavor a nd low in calories. When you eat these - don't need another potato chip!
Similarity: 0.4054 | Review: Excellent product. Exactly what they said it would b e. Superior chamomile from what I have ordered before from other companies. Excel lent value for the price.
Similarity: 0.3856 | Review: We recieved the product on time and in excellent sha pe. It tastes good and meets our expectations. Thanks for the excellent service.
Similarity: 0.384 | Review: Title says it all, a Excellent quality product and Ex cellent service at a fair price. Product came ahead of time and as expected.

Query: poor packaging
Similarity: 0.3246 | Review: I was delighted to find this product online, because I cannot always find it in the stores. However, each package I have opened thus far is at least half reduced to crumbs because of poor packaging f
Similarity: 0.3085 | Review: This review is for the packaging and not the actual mix, since the mix is a very subjective thing and my taste my not match others an d, some wrote about how poor the packaging was. All I can say is wi
Similarity: 0.3084 | Review: The soup is rich and tasty, but DON'T buy it from Am azon. The packaging is so poor that the cans arive dented as did the replacement shipment. I got my money back but now have to pay more to buy this
Similarity: 0.292 | Review: Didnt really enjoy these after wanting to try somethi ng other than aunt jemima's from Dominicks. The pancakes tasted strange and somet imes forgettable, but it might have been due to the poor packaging
Similarity: 0.2811 | Review: was really dissappointed in product... too salty and poor flavor....Convenience is what makes product worth purchasing if you like por k rinds...

Q12: Relevance Analysis

1. 'great product with fast shipping': Retrieves reviews sharing terms related to product quality and shipping.
2. 'disappointed': Retrieves reviews containing matching terms that may indicate customer dissatisfaction.
3. 'excellent taste': Retrieves reviews containing terms related to taste and positive feedback.
4. 'poor packaging': Retrieves reviews containing terms related to packaging and negative feedback.

In [ ]: